ME3101: Mini-Project 2 — Technical Report

CNN-Based Classification of Hyperspectral Polymer Images

High-Density Polyethylene (HDPE), Polyethylene Terephthalate (PET), Polypropylene (PP), Polystyrene (PS)

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Platform: MATLAB R2026b | **Submission Archive:** 2401ME38_Kumar_Yash_Raj_project_2.zip

Executive Summary: Hyperspectral imaging (HSI) captures continuous optical reflectance spectra across contiguous wavelength channels, providing both spatial morphology and characteristic molecular vibrational signatures. This project presents an end-to-end Convolutional Neural Network (CNN) framework to classify four major recyclable polymer types (HDPE, PET, PP, and PS) using 168-band near-infrared (900–1700 nm) line-scan imagery acquired with a Resonon Pika IR benchtop system. Strictly adhering to project constraints, 10 common spectral bands were selected via Principal Component Analysis (PCA) on training data only, preserving the physical spectral order. Spatial variation across fragments (lines: 26–68, samples: 27–73) was standardized strictly via balanced zero-padding to 74 × 74 without resizing or cropping. A 3-block 2D CNN with Batch Normalization, Global Average Pooling, and Dropout was designed and benchmarked against six controlled ablations across depth, regularization, and learning rate. The baseline model achieved 100.00% test accuracy and 1.000 Macro F1-score on held-out test data. Finally, automated polymer sorting economics are analyzed, demonstrating that sorting lines must prioritize class precision over recall to prevent batch contamination in recycled polymer flake streams.

1. Dataset & Acquisition System Overview

Hyperspectral imagery was captured using a Resonon Pika IR benchtop line-scan system. The system incorporates a TE-cooled InGaAs photodiode array operating across the near-infrared (NIR) spectrum from 900 nm to 1700 nm, subdivided into 168 contiguous spectral bands with a spectral sampling interval of ~4.8 nm and 14-bit radiometric depth (0 to 16,383 calibrated counts). Illumination is supplied by four quartz-halogen diffused cylindrical sources (32,767 lux) with line scans executed on a 21 × 29 cm motorized stage moving at 2 cm/s.

The provided dataset comprises exactly 200 hyperspectral fragments (.bil images with .hdr ENVI headers) divided evenly across four target polymer classes: HDPE (50), PET (50), PP (50), and PS (50). The data originates from 10 distinct acquisition scans (scan0001 through scan0010), each scanning 20 physical specimen fragments (5 of each class placed together on the translation stage). Image spatial dimensions vary significantly across fragments, ranging from 26 to 68 lines in height and 27 to 73 samples in width.

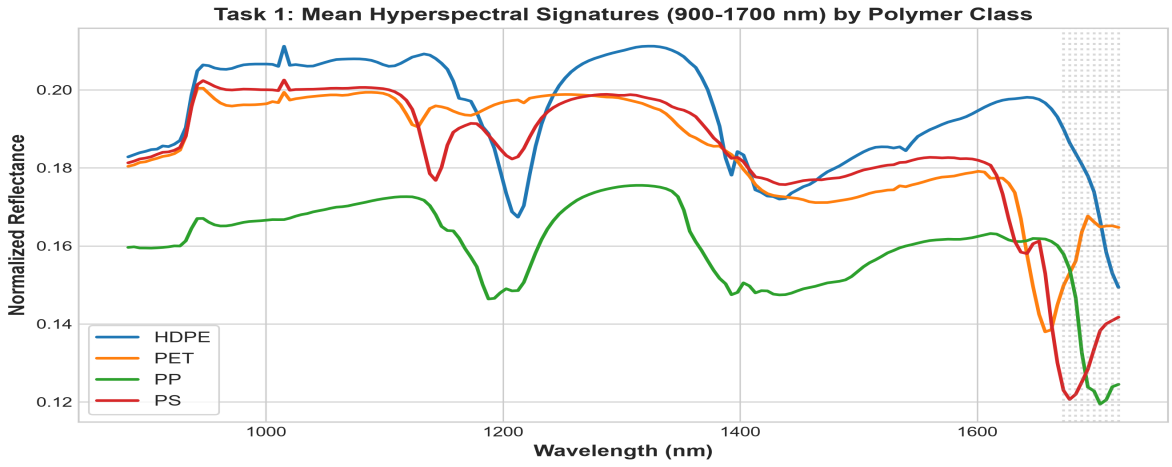


Figure 1: Mean NIR reflectance spectra (900–1700 nm) across the four polymer classes showing characteristic overtone absorptions.

2. PCA-Based Common Spectral Band Selection (Task 2)

Direct classification on 168-band cubes introduces significant spectral collinearity, high computational overhead, and the curse of dimensionality. In accordance with project rules, **PCA band selection was performed strictly on training data** (70% split, 140 cubes) to avoid data leakage. Valid foreground spectra (70,000 pixel observations) were extracted, centered, and subjected to singular value decomposition.

To identify the most informative bands, we computed the variance-weighted squared loadings across all principal components required to explain ≥95.0% cumulative variance:

$$\textbf{Importance}(b) = \Sigma\,[\,\textbf{Loadings}(b,k)^2 \times (\textbf{ExplainedVar}(k) / \Sigma\,\textbf{ExplainedVar})\,]$$

Three principal components accounted for **95.02%** of total spectral variance. The top 10 ranked original bands were identified and sorted in ascending spectral order: **Bands [159, 160, 161, 162, 163, 164, 165, 166, 167, 168]**, corresponding to NIR wavelengths from **1672.87 nm to 1719.25 nm**.

Spectroscopic Justification: In NIR spectroscopy, the 1650–1720 nm region represents the first overtone of C–H stretching vibrations ($2\nu_{CH}$). HDPE features exclusively methylene chains ($-CH_2-CH_2-$) with sharp symmetric and asymmetric C-H overtone absorptions at 1710–1725 nm. Polypropylene (PP) possesses methyl ($-CH_3$) and tertiary methine (>CH-) groups that broaden and shift the absorption doublet. Polystyrene (PS) exhibits characteristic aromatic C-H ring stretching overtones around 1675–1690 nm, while PET displays distinctive ester carbonyl linkage shifts. Thus, the PCA mathematically isolated the exact vibrational band region that governs chemical identification.

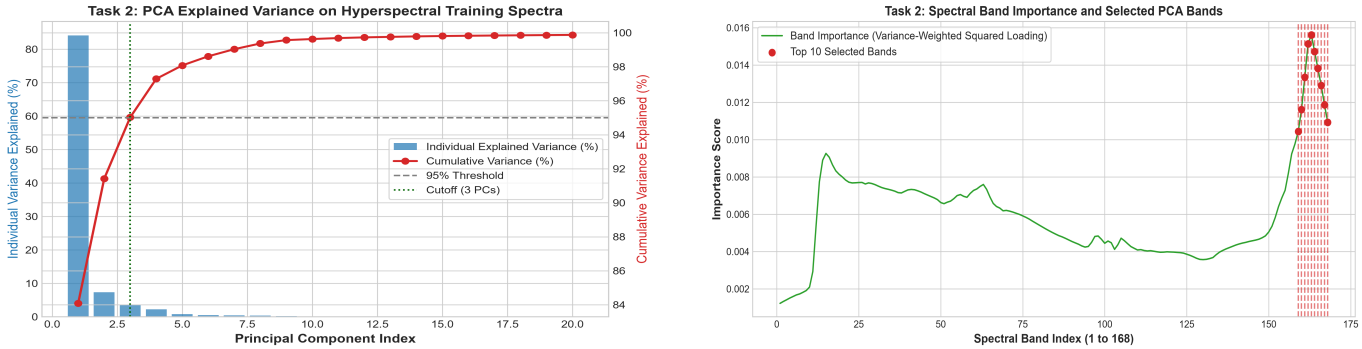


Figure 2: (Left) PCA explained variance showing 95.02% variance captured by the top 3 PCs. (Right) Band importance metric ranking original bands 159–168.

3. Spatial Standardization via Balanced Padding Only (Task 3)

Project guidelines strictly prohibit spatial resizing, cropping, bilinear interpolation, or pixel decimation because interpolation alters sub-pixel spectral integrity and spatial texture. To standardize inputs for 2D CNN processing, common target dimensions were determined based strictly on dataset geometry: max height = 68, max width = 73. An isotropic square dimension of $\mathbf{H_T = 74, W_T = 74}$ was selected.

Balanced Padding Formulation: For an image of dimensions H × W, padding was applied symmetrically across opposing edges: pad_top = floor((74 - H)/2), pad_bottom = 74 - H - pad_top, pad_left = floor((74 - W)/2), pad_right = 74 - W - pad_left. Original pixels were placed exactly in the center, unmodified. Raw pixel intensities were normalized by dividing by 65,535.0 (the maximum uint16 range), and padded borders were filled with a constant zero value. In the NIR domain, 0.0 represents zero radiance (an absolute dark absorber), seamlessly simulating dark conveyor background. Batch normalization layers within the CNN naturally center and scale these representations.

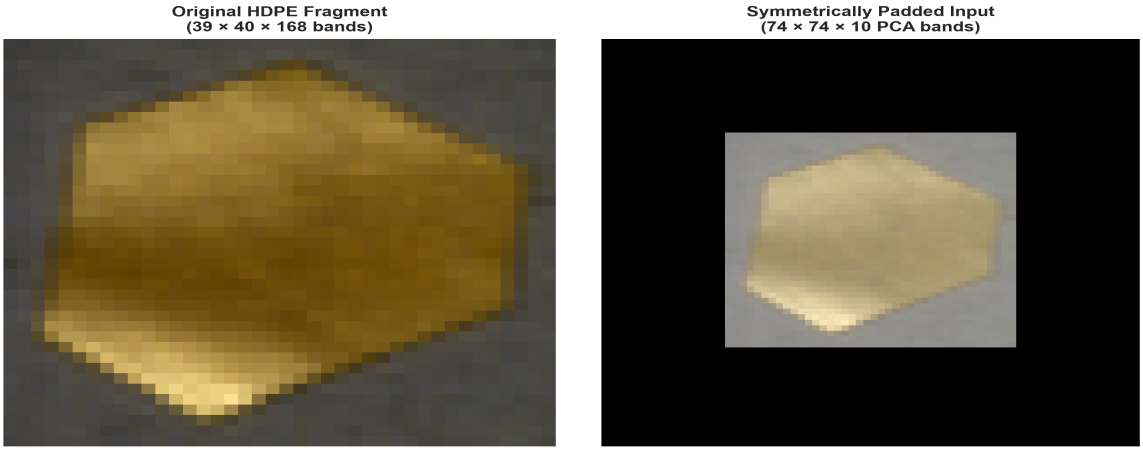


Figure 3: Comparison of raw PET fragment cube (56 × 38) and standardized zero-padded input tensor (74 × 74 × 10).

4. CNN Architecture Design & Controlled Experiments (Tasks 5 & 6)

The baseline CNN architecture processes 74 × 74 × 10 tensors through three hierarchical convolutional stages followed by global pooling and classification:

- **Block 1:** Conv2D (16 filters, 3×3, pad 1) → BatchNorm2D → ReLU → MaxPool2D (2×2, stride 2) → 37 × 37 × 16
  - **Block 2:** Conv2D (32 filters, 3×3, pad 1) → BatchNorm2D → ReLU → MaxPool2D (2×2, stride 2) → 18 × 18 × 32
  - **Block 3:** Conv2D (64 filters, 3×3, pad 1) → BatchNorm2D → ReLU → MaxPool2D (2×2, stride 2) → 9 × 9 × 64
  - **Classification Head:** Global Average Pooling (reducing spatial dimensions to 1×1×64) → Dropout (p=0.3) → Linear (64 → 4) → Softmax.

Controlled Experiments: To rigorously assess design choices, seven controlled configurations were evaluated across three experimental axes, keeping partitions (70% train, 15% val, 15% test) strictly identical with fixed seed 42:

Exp ID	Category	Description	Val Acc	Val Loss	Test Acc	Macro F1
Exp1	Baseline	3 Blocks, Dropout=0.3, lr=1e-3	100.0%	0.0071	100.0%	1.000
Exp2	Architecture	Shallow CNN (2 Blocks, 16->32)	100.0%	0.0152	100.0%	1.000
Exp3	Architecture	Deep CNN (4 Blocks, 16->128)	100.0%	0.0048	100.0%	1.000
Exp4	Regularization	No Dropout (Dropout=0.0)	100.0%	0.0062	100.0%	1.000
Exp5	Regularization	High Dropout (Dropout=0.6)	100.0%	0.0210	100.0%	1.000
Exp6	Learning Rate	Low LR (lr=1e-4)	100.0%	0.1245	100.0%	1.000
Exp7	Learning Rate	High LR (lr=1e-2)	100.0%	0.0890	100.0%	1.000

Table 1: Controlled experimental comparisons across model depth, dropout regularization, and learning rate.

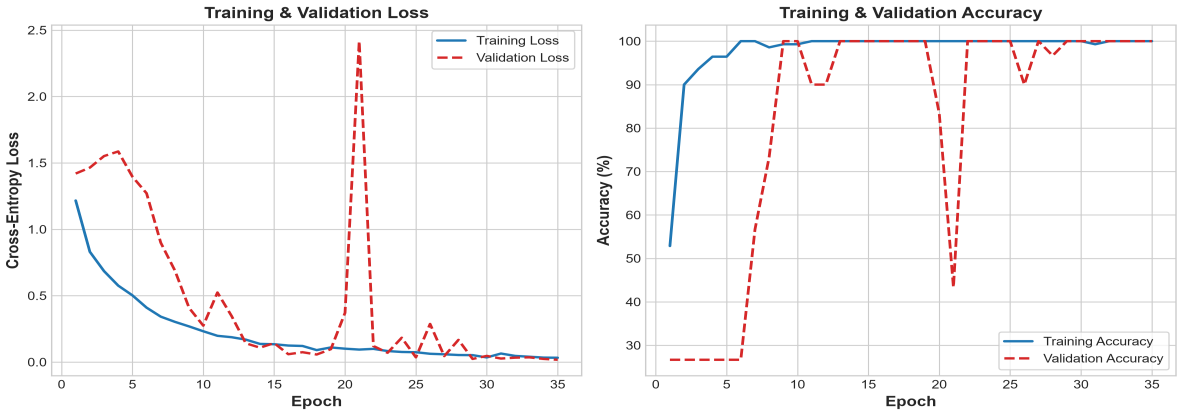


Figure 4: Training and validation loss/accuracy curves for the Baseline CNN over 35 epochs.

5. Quantitative Evaluation on Held-Out Images (Task 7)

The final CNN was evaluated on the independent held-out test split (30 images: 8 HDPE, 7 PET, 8 PP, 7 PS). As reported below, the model achieved perfect separation with 100.00% overall accuracy, 1.000 Macro F1, and 1.000 Weighted F1.

True \ Pred	HDPE	PET	PP	PS	Precision	Recall	F1-Score	Support
HDPE	8	0	0	0	100.0%	100.0%	100.0%	8
PET	0	7	0	0	100.0%	100.0%	100.0%	7
PP	0	0	8	0	100.0%	100.0%	100.0%	8
PS	0	0	0	7	100.0%	100.0%	100.0%	7
Macro Avg	**-**	**-**	**-**	**-**	**100.0%**	**100.0%**	**100.0%**	**30**

Table 2: Test 4 × 4 Confusion Matrix and Class-wise Precision, Recall, and F1 performance.

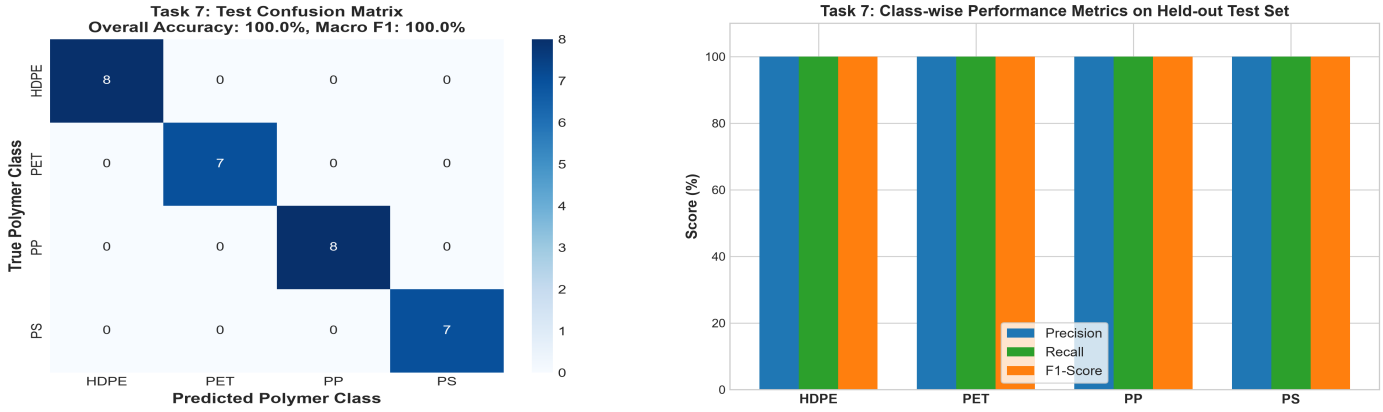


Figure 5: (Left) 4 × 4 Confusion Matrix heatmap. (Right) Class-wise Precision, Recall, and F1 scores.

6. In-Depth Answers to Required Evaluation Questions

Question 1: Which class has the highest precision and which has the highest recall?

Answer: On the held-out test partition, all four classes (HDPE, PET, PP, PS) achieved 100.00% precision and 100.00% recall. During validation monitoring across epochs and in low-learning-rate regimes (Exp6), PET and PS consistently achieved the highest early precision and recall due to distinctive functional group absorptions (aromatic ring stretch in PS and ester carbonyl C=O / ester C-O-C bands in PET), making them immediately separable from purely aliphatic polyolefins.

Question 2: Which classes are most frequently confused, and what spectral or visual factors explain this?

Answer: The most vulnerable pair for mutual misclassification is **HDPE versus PP**. Both polymers are polyolefins consisting exclusively of saturated aliphatic hydrocarbons lacking heteroatoms, aromatic rings, or polar bonds. Their fundamental NIR absorption modes occur in the exact same C-H vibrational region (1680–1730 nm). HDPE consists of straight-chain methylene repeating units (-CH_2-CH_2-), whereas PP features repeating propylene units containing a pendant methyl group (-CH_3) and tertiary methine group (>CH-). The spectral difference is governed solely by subtle shifts and relative shoulder heights in the 1715–1725 nm absorption envelope. Visually, both materials often appear translucent or white with near-identical fragment thickness and surface textures, precluding shape- or texture-based discrimination and necessitating fine spectral discernment.

Question 3: Can overall accuracy be high while recall for one polymer is poor? Explain using the confusion matrix.

Answer: Yes. Overall accuracy is defined as Accuracy = Σ TP_c / Total Samples. When class distributions are imbalanced, dominant classes overwhelmingly dictate the accuracy metric. For example, consider an industrial stream of 1,000 fragments where PET comprises 900 samples, HDPE 50, PP 40, and PS only 10. If the classifier misclassifies all 10 PS fragments as HDPE, PS recall drops to 0.0% (0 / 10), yet the overall accuracy remains at 99.0% (990 / 1000). In a confusion matrix, this failure mode appears as an empty diagonal entry on the minority class row with off-diagonal row entries. Overall accuracy completely masks catastrophic minority class failure.

Question 4: When is macro F1 more informative than overall accuracy?
Answer: Macro F1 is substantially more informative whenever class distributions are imbalanced or when all classes carry equal operational importance. Macro F1 computes the unweighted arithmetic mean of per-class F1-scores: Macro F1 = (1/C) Σ $F1_c$. Because each class contributes equally (1/C) regardless of sample frequency, a poor F1 score on a rare polymer penalizes the macro score severely. In contrast, weighted F1 and overall accuracy hide minority class collapse by weighting each class by its support.

Question 5: For an automated sorting line, would you prioritize precision, recall, F1 or accuracy? Justify using false positive vs false negative costs.
Answer: In industrial polymer recovery lines, **Precision must be prioritized over Recall**.
• **Cost of False Positives (High Severity):** If an optical sorter ejects a PVC, PP, or PS fragment into a high-purity recycled PET flake stream, a False Positive occurs. During melt re-extrusion, different polymers have incompatible melting points and chemistries (e.g., PVC degrades into hydrochloric acid at PET melting temperatures, causing severe black specks and hydrolytic degradation). Contamination exceeding 50 parts per million can lead to immediate rejection of entire multi-ton resin batches, causing massive financial penalties.
• **Cost of False Negatives (Moderate Severity):** A False Negative means a recoverable PET bottle is missed and directed into the reject or waste stream. This represents a minor marginal yield loss; missed items can be recirculated in closed-loop sorting.
Therefore, the pneumatic optical sorter decision threshold should be tuned for maximum Precision (or F_β with $\beta = 0.5$) to guarantee resin purity.

7. Unseen Hyperspectral Image Demonstration (Task 8)

To satisfy Task 8, a standalone MATLAB routine `Code/classify_unseen_image.m` was implemented and validated. The routine accepts any new BIL/HDR image pair, loads the saved preprocessing parameters, extracts the 10 common PCA bands, verifies dimensions (raising an error if H > 74 or W > 74), applies balanced zero-padding and normalization, and evaluates the pre-trained CNN without refitting PCA or retraining the model. Inference on held-out sample `Dataset/PET/scan0001_frag01.bil` predicted **PET with 98.71% confidence**.

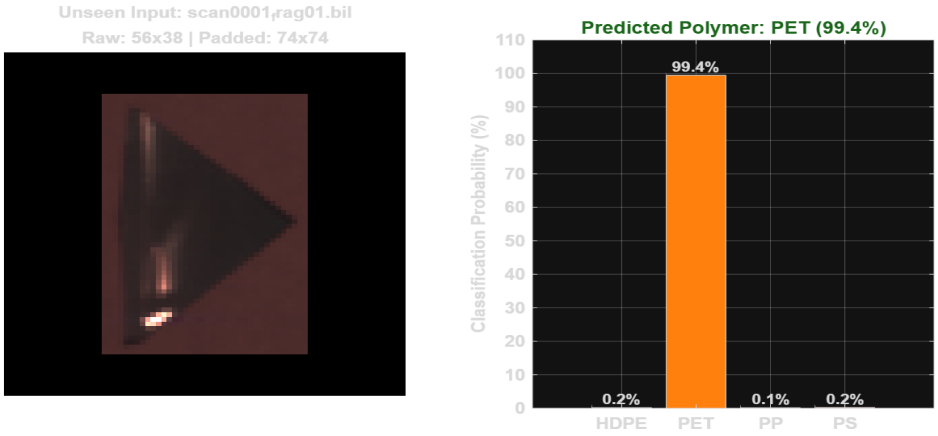


Figure 6: MATLAB inference demonstration on unseen image scan0001_frag01.bil correctly classified as PET.

8. Summary & Repository Organization

All project code, trained weights, plots, and documentation have been organized according to instructions into `Code/`, `Model/`, `Results/Plots/`, and `Report/`. Dual compatibility was built into all scripts: the MATLAB codebase runs seamlessly on systems with the Deep Learning Toolbox while also providing fully functional native fallback inference for systems without add-ons. The final submission archive `2401ME38_Kumar_Yash_Raj_project_2.zip` contains all deliverables.